BPS PROVINSI JAWA TENGAH · 14 AUGUST 2026 · SEMARANG

Digitalising Social Protection: Better Targeting and Delivery

Arief Anshory Yusuf

▸ BIO

Social assistance in Indonesia misses a large share of the households it is meant for. This asks three questions: how large the gap is, where it comes from, and which parts of it digitalisation can actually close.

THE PROBLEM

We cannot see who is poor. We only see signs.

A hunter cannot see the animal in the tree. He can only see the leaves move. Targeting works the same way: welfare itself is never observed, only proxies for it.

Who decides

Tree A hides the **musang** — the household that qualifies. Tree B holds **birds** — households that look the same from outside.

Exclusion error — the musang escapes: a household that qualifies gets nothing

Inclusion error — a bird is hit: a household that does not qualify receives the benefit

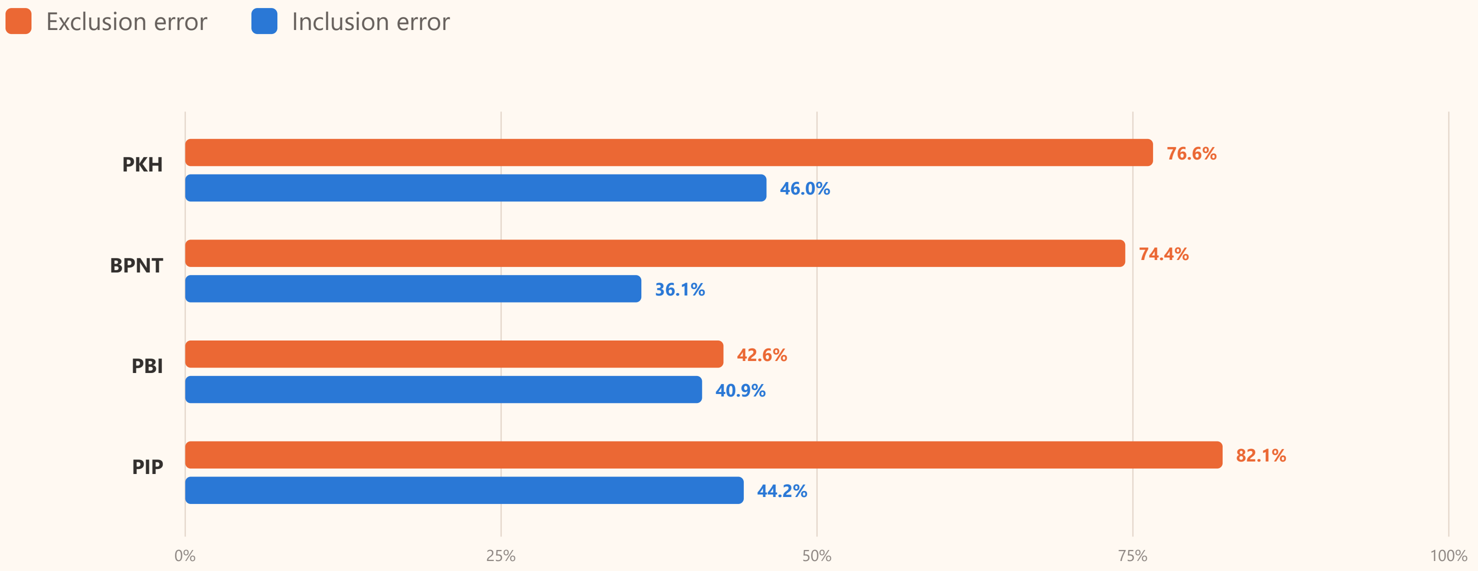
I How well does it reach people?

Four programmes, measured against the households and people they are meant to serve.

FIGURE 1

How much of the target group is missed

Susenas March 2025. Exclusion error is the share of the target group that receives nothing. Inclusion error is the share of recipients who fall outside the target group.



How the target group is defined

PKH deciles 1–4, and the household must meet the programme conditions (a child under 5, a school-age member, or an elderly member).

BPNT deciles 1–5. **PBI** deciles 1–5, counted per person. **PIP** deciles 1–4, school age 6–27, counted per person.

Welfare is real expenditure per person, adjusted for price differences between provinces and between urban and rural areas.

The cut-off matters

Move the PKH line to deciles 1–3 and inclusion error becomes **57.7%**. Move it to 1–5 and it becomes **35.7%**. Nothing about the programme changed.

FIGURE 2

Three years: two programmes improving, one not

Exclusion error, March 2023 to March 2025, on the same definitions. PBI and PIP reach more of their target group each year. PKH does not.

Exclusion error — eligible households not reached

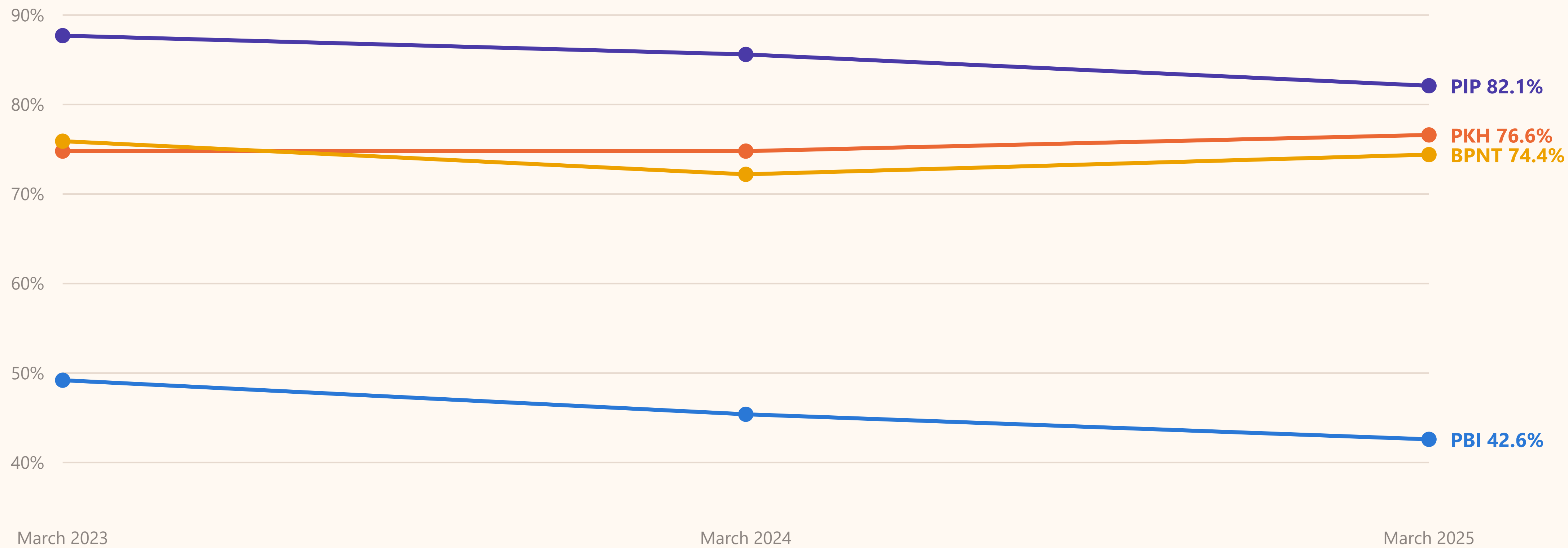
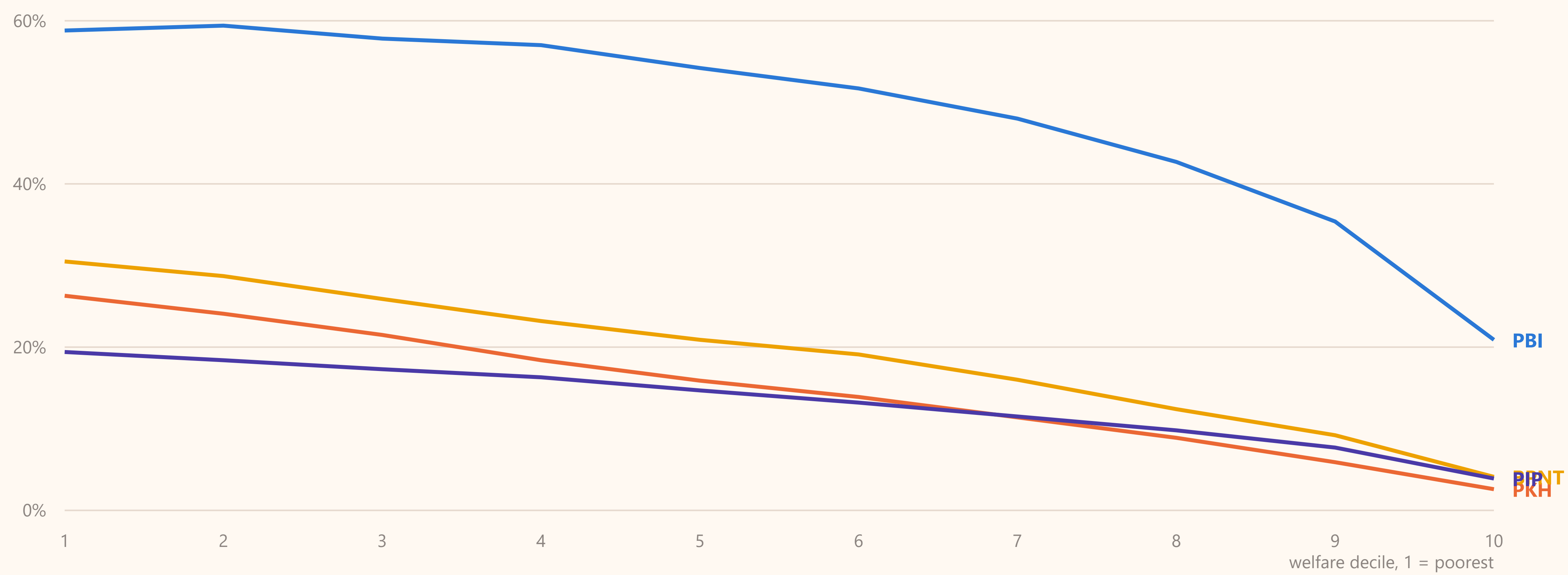


FIGURE 3

The ranking works. The coverage does not.

Share of households or people receiving each programme, by welfare decile. Coverage falls steadily from the poorest decile to the richest in every programme, so the ranking is pointing the right way.

Share receiving the programme



TWO MEASURES

Why BPS reports 0.06% and this survey shows 46%

Both numbers are correct. They answer different questions.

BPS — 0.06%

11,014 recipients out of 18.15 million, reported 13 April 2026.

This counts recipients who **became ineligible between one quarterly DTSEN update and the next**. It is a measure of movement: how many changed status this quarter.

Survey — 46%

Susenas March 2025, PKH.

This counts recipients who **sit outside the target group at a single point in time**. It is a measure of the standing position: who is on the list today.

A quarterly change of 0.06% and a standing position of 46% can both be true at once. The first says the list is stable. The second says where the list started.

FIGURE 4

Most of the gap is the budget, not the data

PKH has about 24 million eligible households and funds about 10 to 11 million places. Even with perfect targeting, most eligible households would still receive nothing.

~24 million

households meet the PKH criteria

10–11 million

places the budget funds

18 million

keluarga: the legal cap on bansos recipients, set as a quota rather than by decile (Permensos 3/2025 Pasal 17)

What follows

Being in decile 1 does not entitle a household to anything. The rules filter by criteria first, then cut the list down to the funded quota.

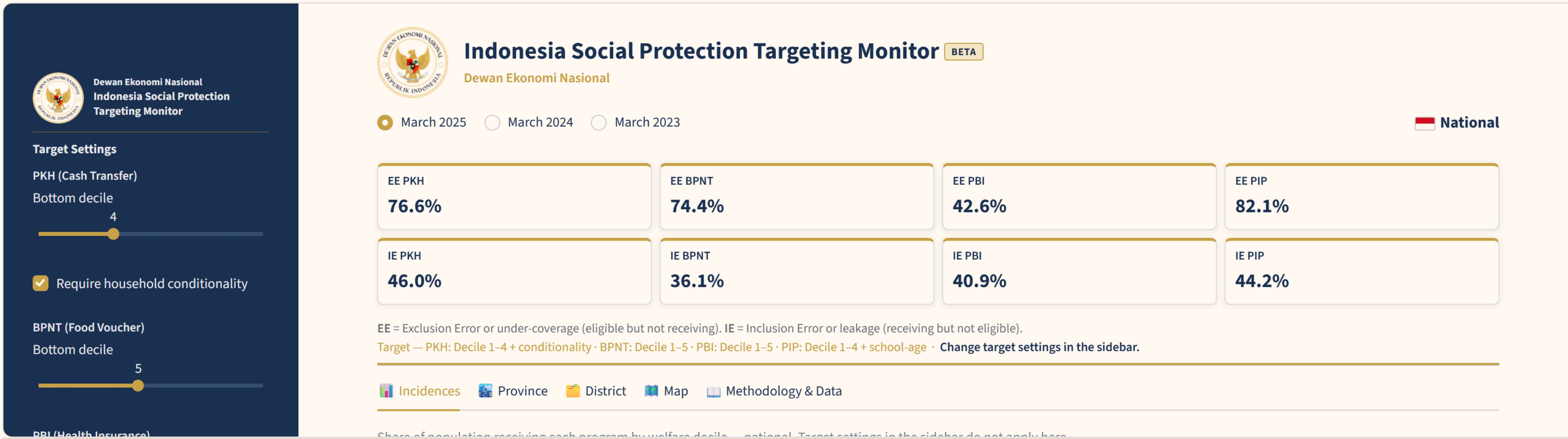
So a large part of the 76.6% exclusion error is a budget decision that no data system can undo. Better data changes **who** receives the benefit. It does not change **how many**.

This also explains Figure 2: PBI, which covers about half the population, improved fastest. PKH, which is capped, did not.

THE DATA

These numbers are public and can be checked

The Dewan Ekonomi Nasional targeting monitor computes exclusion and inclusion error directly from Susenas microdata, for three years and for every province and district. Anyone can change the target definition and see the numbers move.



II Is this unusual?

Other countries publish the same measures. What their experience shows is where the error actually comes from.

FIGURE 5

The same range everywhere

Exclusion and inclusion error for social assistance programmes with published figures. Exclusion error runs from about 50% to 90%, inclusion error from about 24% to 70%. Indonesia sits inside both ranges.

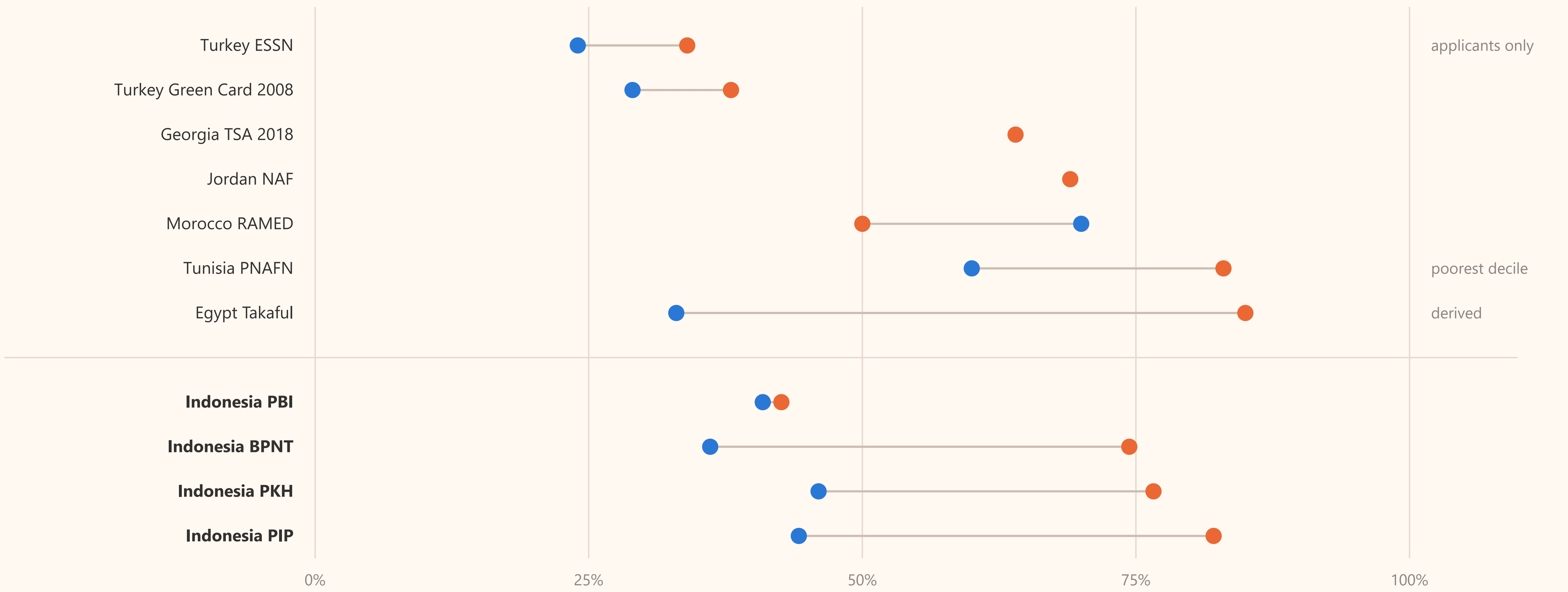
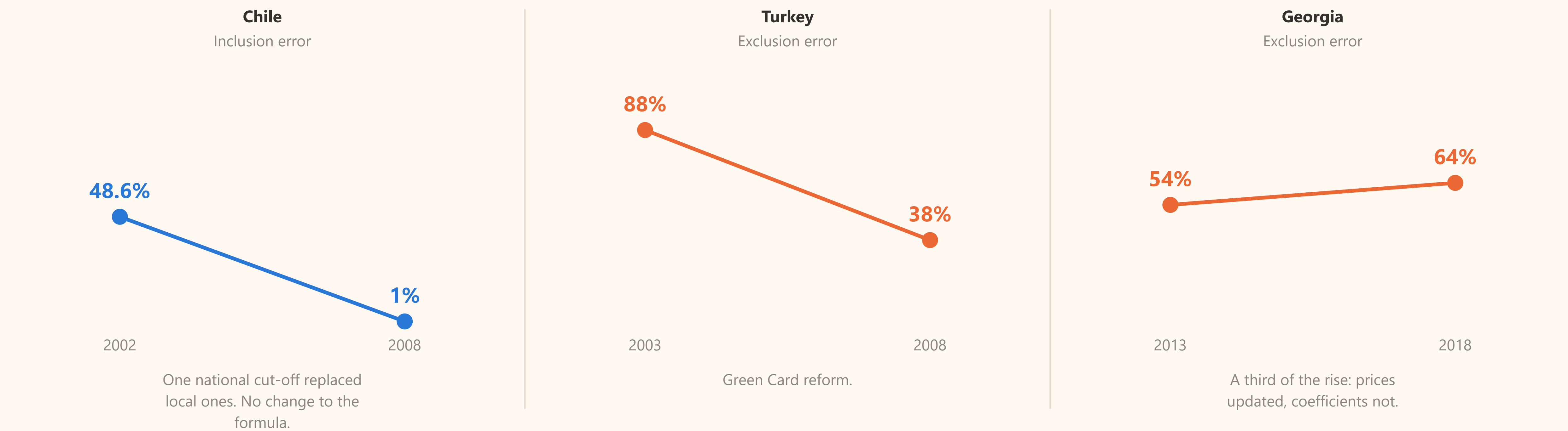


FIGURE 6

What actually changed the numbers elsewhere

In each case where measured error fell sharply, what changed was a rule, a threshold or a process. In none of them was it a better statistical model.



Egypt is the clearest case. Only half of the poorest fifth of households ever applied for Takaful. Researchers calculated that the formula on its own would have admitted 46% non-poor recipients; the figure actually observed was 33%. The programme was more accurate than its formula, because the people who did not need it largely did not apply. The exclusion problem was in **who came forward**, not in the scoring.

III Why is it imperfect?

Two reasons. One is a limit on what any formula can know. The other is everything that happens after the formula.

FIGURE 7

What a country can verify sets what it can do

Targeting methods form a ladder. Which rung a country can stand on is set by how much household income the government can verify from administrative records — not by the quality of its statisticians.

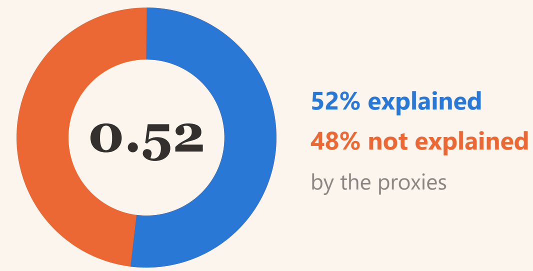
Verified means test
Income read directly from records
Needs: Most income visible in records
— Chile

Hybrid means test
Verified income plus estimated informal income
Needs: Tax, payroll and pension records linked
— Turkey — the next step

Advanced proxy means test
Proxies, recalibrated and cross-checked
Needs: Regular survey plus administrative checks

Proxy means test
Welfare estimated from observable proxies
Needs: A household survey
— Indonesia is here

Half of what matters is invisible to the formula



Indonesia's own PKH proxy-means formulas had an average R^2 of **0.52**. About half the variation in household consumption is explained by the observable proxies. The other half is not.

Why Indonesia cannot copy Chile

Chile can rank households on recorded income. Recorded income is never higher than true income, so that filter can place a household too low but never too high — it cannot wrongly exclude the poor. It changes the question from *who is poor?* to *who is clearly not poor?*

That requires most income to be visible in records. In Indonesia, formal employment is under 20% of the workforce. Turkey, not Chile, is the next step.

FIGURE 8

Five sources of error. Only two are the formula.

Errors arise at two separate stages, and each stage produces both kinds of error. In the outcome data they look identical.

Choosing who to help — pensasaran

Delivering the help — penyaluran

1

Out-of-date register

The household's circumstances changed after it was last recorded.

TARGETING

2

Inaccurate formula

The proxies do not capture the household's real situation.

TARGETING

3

Budget limits

The list is cut to the funded quota, whatever the criteria say.

DELIVERY

4

Business process

Records enter unchecked, appeals go unanswered, payments are delayed.

DELIVERY

5

Elite capture

Local influence decides who appears on the list.

DELIVERY

IV What can digitalisation change?

Registration, verification and payment can be improved now. What the formula can know cannot.

FIGURE 9

Self-registration on its own does not work

No country has succeeded with self-service registration alone. Every system where opening registration improved targeting added four things alongside it.

1

Staff to help people apply

19 of 20 on-demand social registries receive applications at staffed local offices. In Surabaya, **84%** of registrations came through an agent and only 16% were self-registered.

INDONESIA: YES

2

Going out to find people

Brazil ran open registration for five years and reached 22 million families, and still found **over one million** more extremely poor families in a single year once it went looking for them.

INDONESIA: PILOTS ONLY

3

Appeals that get answered

An appeal channel corrects the households the formula missed — but only if the appeals are actually decided.

INDONESIA: PARTLY

4

Checking records before accepting them

Administrative data checked at the moment of application, before the record enters the register.

INDONESIA: NO

Opening registration does help. A randomised trial in 400 Indonesian villages found that when households had to come forward and apply, the people who received benefits were **18 to 19% poorer** than under automatic enrolment, and the very poorest were twice as likely to be reached. But only about **60%** of the very poorest applied — and making the process harder screened out poor and rich equally, so any extra difficulty is a cost borne by the poor for no gain.

FIGURE 10

Indonesia's own audit found both ends open

The state audit board examined the register that preceded DTSEN. Records were being accepted without checking, and appeals were being closed without a decision.

224,635

new applications recorded automatically **without verification** — 15,970 of them without the required photographs. BPK: this creates inclusion error.

7,267

appeals closed automatically without being decided — produces exclusion error. Of these, **735 in Jawa Tengah**, 2,085 in Jawa Barat, 758 in Sumatera Utara.

Turkey's answer is simple

In Turkey's integrated system, when an application contains missing or inconsistent information, an alert appears and **the application is frozen until the information is validated**. The applicant cannot proceed and the record cannot enter the register.

Administrative records behind it are refreshed at least every 45 days, and supply about 40% of the information used to score a household. The remaining 60% still comes from a home visit each year.

This is one transferable design feature, not a whole system: **check the record before you accept it, not after.**

CONCLUSION

What digitalisation can and cannot fix

Of the five sources of error, the two that digitalisation closes fastest are not the statistical formula.

SOURCE OF ERROR	CAN DIGITALISATION FIX IT?	
Out-of-date register	**Yes, directly**	Continuous registration and administrative updates keep the record current.
Business process	**Yes, and fastest**	Check records at the point of application; decide appeals; pay without delay.
Elite capture	**Partly**	An audit trail and published lists make local discretion visible.
Inaccurate formula	**Only up to a limit**	Bounded by how much income can be verified. It improves as formal employment grows, not as the model improves.
Budget limits	**No**	Not a data problem. Better data changes who receives, not how many.

The digital public infrastructure described earlier today is necessary. This is a map of what it will and will not fix.

Sources. Susenas March 2023, 2024 and 2025, own calculation following the definitions of the DEN Indonesia Social Protection Targeting Monitor. Cross-country figures from World Bank and IFPRI publications, each with its own definition of the target population — see the note under each figure. Framework and illustrations from *Salah Sasaran* (Mei 2026). Audit findings from BPK RI, IHPS II 2024.